

GUNEYKAN OZGUL

New York City, NY, USA
guneykanozgul@gmail.com

EDUCATION

Pennsylvania State University, PA *2021-2025*

Ph.D. in Computer Science & Engineering

Dissertation: Quantum Algorithms For Markov Chain Methods In Machine Learning and Optimization

Supervised by Chunhao Wang, Mehrdad Mahdavi

Overall GPA: 3.88

Boston University, MA *2018-2019*

Graduate Student in Physics

Theoretical Particle Physics and Cosmology,

Supervised by Martin Schmaltz

No degree obtained

Overall GPA: 3.90

Bogazici University, Istanbul *2012-2018*

B.S in Physics (Double Major)

Overall GPA: 3.79 (High honor)

Bogazici University, Istanbul *2012-2018*

B.S in Computer Engineering

Senior Project: A Hybrid Recommendation System with Application on Sparse Twitter Data

Supervised by Ali Taylan Cemgil

Overall GPA: 3.79 (High honor, graduated 2nd in class)

PUBLICATIONS AND PREPRINTS

- **Ozgul, G.**, Li, X., Mahdavi, M., and Wang, C. Quantum speedups for monte carlo markov chain methods with application to optimization, 2025. Preprint. In submission to ICLR 2026.
- Chakrabarti, S., Herman, D., **Ozgul, G.**, Zhu, S., Augustino, B., Hao, T., He, Z., Shaydulin, R., and Pistoia, M. Generalized short path algorithms: Towards super-quadratic speedup over markov chain search for combinatorial optimization, 2024 [Accepted TQC].Preprint. URL <https://arxiv.org/abs/2410.23270>.
- **Ozgul, G.**, Li, X., Mahdavi, M., and Wang, C. Stochastic quantum sampling for non-logconcave distributions and estimating partition functions. Proceedings of the 41st International Conference on Machine Learning, volume 235 of Proceedings of Machine Learning Research, pp. 38953–38982. PMLR, 21–27 Jul 2024. URL <https://proceedings.mlr.press/v235/ozgul24a.html>.

WORK EXPERIENCE

JPMorgan & Chase, NYC *Current*

Quantum Computing Associate Research Scientist

JPMorgan & Chase, NYC *Summer 2024*

Quantum Computing Intern

Pennsylvania State University, PA *2021-2025*

Graduate Researcher

Boston University, MA

Teaching Assistant

2018-2019

Pointr, Istanbul

Research and Development Engineer

2019-2021

TEACHING EXPERIENCE

Pennsylvania State University

Teaching Assistant

2021-2023

State College, PA

Fall 2021 - CMPSC 497 (Introduction to Quantum Computing)

Spring 2022 - CMPSC 448 (Machine Learning and Algorithmic AI)

Fall 2022 - CMPSC 497 (Introduction to Quantum Computing)

Spring 2023 - CMPSC 448 (Machine Learning and Algorithmic AI)

Boston University

Teaching Assistant

2018-2019

Boston, MA

Fall 2018 - PY 211 (Intermediate Electromagnetism)

Spring 2019 - PY 106 (Elementary Electromagnetism)

Summer 2019-1 - PY 212 (Intermediate Mechanics)

Summer 2019-2 - PY 211 (Intermediate Electromagnetism)

OTHER PROJECTS AND ROLES

Inzva, AI Projects 4

Project Lead

2020

Istanbul, TR

Text Classification using Graph Neural Networks

Huawei

Trainee on 5G Technology

2017

Shenzen, China

Selected by Huawei training in Beijing and Shenzen

Bogazici University

Undergraduate Researcher

2017

Istanbul, TR

Variance Reduction in Multidimensional Monte Carlo Integration

Under supervision of Prof. Taylan Akdogan

TECHNICAL SKILLS

Languages

Turkish (Native), English (Advanced; TOEFL: 110/120), Spanish (Beginner)

Computer Languages

(Advanced) C/C++, Python, JAVA, Matlab, Mathematica

Tools

Git, LATEX

Frameworks

Qt, Unity, Tensorflow, Pytorch